



InfiniteCooling

Midwest Power Plant Saves \$500K Annually with TowerPulse™

The Midwest power plant, a 620 MW combined-cycle facility, plays a key role in supplying reliable energy to the regional grid. Operating as a base-loaded unit with capacity factors around 80%, it relies on its cooling towers to maintain efficiency, particularly during peak load. These induced-draft SPX cooling towers, consisting of 12 cells with 34-foot fans, are essential for optimal performance under varying weather conditions.

The towers use treated gray water from a local sewage treatment plant, consuming 2 to 3 million gallons daily. With sustainability goals in mind, the plant faced pressure to reduce water usage while maintaining efficiency. However, the cooling towers' condition and their impact on overall performance had not been fully assessed, raising concerns about potential inefficiencies.

To address this, the plant partnered with Infinite Cooling to implement the TowerPulse™ system, leveraging its expertise in optimizing cooling tower operations, improving efficiency, extending equipment lifespan, and reducing water consumption.

The Challenge

The Midwest power plant faced several challenges with its cooling tower system:

1. **Unknown Operational Condition:** The cooling towers' performance was largely unmonitored, leaving operators without insights into inefficiencies affecting plant performance.
2. **Impact on Production Efficiency:** Any drop in cooling tower performance raised condenser backpressure, reducing steam turbine output. Since the turbine contributes 40% of the plant's production, inefficiencies directly impacted electricity generation.

3. **High Water Consumption:** With increasing environmental and regulatory pressures, the plant needed to reduce water use while maintaining cooling. Though treated sewage water reduced fresh water demand, further reductions were necessary. Using treated gray water also increased fouling risks and complicated water treatment, requiring more maintenance and chemicals to maintain system performance.

Balancing these operational and environmental demands was a key priority for management.

The Solution

To address these challenges, Infinite Cooling partnered with the plant management to install its TowerPulse™ package, featuring wireless sensors and advanced algorithms via user-friendly software.

The TowerPulse™ system was installed within 1.5 weeks, fully operational without interrupting plant operations. Sensors were added to monitor ambient temperature, humidity, water flows, water temperatures, and fan RPM across all 12 cooling tower cells. After installation, the system was connected to the cloud, and operators began receiving real-time metrics on cooling tower health, optimization opportunities, and anomaly detection.

Cooling Tower Health

Using the TowerPulse™ dashboard, operators monitored cooling tower performance in real time. Key performance indicators revealed the tower was operating at 80% capability (a measure of efficiency), with a cooling water temperature 2°C higher than the target. This inefficiency was estimated to cost the plant an average 0.8MW of lost power generation. The system identified a likely fan issue and recommended maintenance to improve airflow and restore efficiency.

Process Optimization Opportunities

Another parameter that was monitored was the condenser performance through continuous monitoring of condenser pressure, steam turbine output, and cooling tower parameters. One of the provided metrics was real-time NTU (Number of Transfer Units) data, which is a measure of how well the condenser is transferring heat from the steam. TowerPulse™ algorithms correlated the condenser performance with the cooling tower water treatment program and recommended process improvement measures. In one instance, the system recommended an increase in frequency of flushing, where water flow is increased and

chemicals added to wash off the system from biological growth and scale. The plant was briefly increasing flowrate and adding chemicals every two weeks to mitigate biofouling. However, TowerPulse™ measured that after a week, the condenser performance would deteriorate to a point where the plant was losing an average 1MW of production due to biofouling on the condenser. Production would only return to nominal levels after a flush. The 2-week frequency was chosen based on “rules of thumb” but with direct measurements, the process frequency was adjusted to weekly to reflect the reality of the system and increase production.

Anomaly Detection

One of the issues that plagues cooling towers is fan motor and gearbox failures, which not only cost money in equipment and labor for replacement but also shut down the affected cells for extended periods and result in reduced production during that time.

The plant was using a simple algorithm to determine the number of fans that should be on at any given time, based on a cold water temperature setpoint. While the control philosophy was sound, TowerPulse analysis discovered that the implementation was causing equipment damage. The algorithm was triggering rapid speed changes with numerous repeated hot starts, which was imposing significant wear on the mechanical equipment with little benefit to energy usage in the tower. An updated control scheme was suggested to smooth the operational profile and reduce maintenance frequency.



The Results

Over the trailing year, Infinite Cooling's TowerPulse™ technology enabled significant enhancements to the power plant's induced draft cooling towers, optimized performance, and lower maintenance needs.

The facility's annual increased profits are estimated at over \$500,000 from production increases driven by cooling tower improvements and process optimization. Some of the drivers in this first year of operation were the identification of fan maintenance needs that will increase airflow, the optimization of the water treatment program, and better water flow control. Since these improvements are driven by turbine efficiency, the additional production did not involve more fuel but rather enabled more generation from the same amount of fuel. Estimated annual energy savings are 10 million kWh, corresponding to 7,000 metric tons of avoided CO2 emissions, further contributing to the plant's goals to provide the cleanest possible power. Water consumption in the cooling tower was also reduced due to the increased system efficiency with an estimated 4 million gallons per year of savings, reducing the reliance on scarce local water resources.

In addition to production increases, optimized fan controls were able to reduce wear and tear on mechanical equipment with estimated annual savings of over \$100,000. More generally, TowerPulse allowed the plant to move from time-based maintenance to condition-based maintenance where only real issues are addressed, and they are addressed promptly.

Finally, the tool generated precise data that was used in a study to quantify the ROI of a VFD retrofit project for fan motors. While the project has not yet been implemented, TowerPulse™ calculated the estimated savings with VFDs and showed that it would have a short ROI.

With these savings, TowerPulse™ has become a valuable tool in the hands of plant operators and engineers to maximize production, reliability, and sustainability.

Annual Impact

Energy Savings

10,000 MWh

Water Savings

4 million gallons of high-purity water

Cost Savings

\$500,000+ annually

CO2 Emissions Reduction

7,000 metric tons

About Infinite Cooling

Infinite Cooling is an award-winning technology company dedicated to enabling efficient and sustainable cooling. The company's patented solutions, including TowerPulse™ and WaterPanel™, optimize the efficiency of cooling systems, significantly reduce water and energy consumption, and lower operating costs. Co-founded by MIT researchers Dr. Maher Damak, Dr. Karim Khalil, and Prof. Kripa Varanasi, Infinite Cooling is the global leader in enhancing industrial cooling across the globe. The company has received numerous accolades from the Department of Energy, Edison Awards, MIT, and MassChallenge. For more information, visit [Infinite Cooling](https://infinite-cooling.com) or contact info@infinite-cooling.com.